

Corrections to the *Omega at g^2* Draft

Located errors, internal inconsistencies and conceptual gaps in
Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order
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Method. Every item gives the equation number and the specific object. Section A lists errors verified by explicit re-derivation from the paper's own matrix elements (D.1)–(D.7) and its own conventions. Section B lists internal contradictions between two places in the paper, and says which one I believe is right. Section C is conceptual; section D is typography and prose. What was checked and found *correct* is listed first, so the verified ground is explicit.

The one that breaks a stated result. Eq. (6.4) prints the two A-terms with opposite signs, making the bracket anti-Hermitian — so it cannot cancel the Hermitian H_g , and Eq. (6.5) does not follow. Two independent arguments fix the sign as $+A(p^2/2p^+)a^\dagger$.

A coefficient of Ω is wrong. Eq. (4.55) is missing the factor i from the colour-charge commutator and the factor $1/\sqrt{(k^+p^+)}$. The error propagates into Eq. (5.4), where the two terms inside one bracket then differ dimensionally.

Also located. Eq. (4.36) omits the instantaneous piece that Eq. (4.35) promises and Eq. (5.3) actually uses; a stray momentum k and a swapped $p \leftrightarrow q$ assignment in Eq. (5.3); colour factor and p argument in Eq. (4.86); g^4 for g^2 in Eq. (4.94); unbalanced brackets in Eq. (2.23); a $(2\pi)^9$ measure mismatch between Eqs. (4.97) and (4.99); and two duplicate bibliography entries.

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Review of $\Omega_{\text{at}_g^2}$ (Only gluons) DRAFT — located errors

Draft reviewed: 57 pp., "Soft Gluon Wave Function and Evolution Operator in the CGC at Next-to-Leading Order", prepared for JHEP.

Every item below gives the equation number and the specific object. Items in **§A** I verified by explicit re-derivation from the paper's own matrix elements (D.1)–(D.7) and its own conventions; they are errors. Items in **§B** are internal contradictions between two places in the paper — one of the two must change, and I say which I believe is right. **§C** is conceptual/argumentative. **§D** is typography and prose.

Conventions I verified first, so that everything downstream rests on checked ground:

- **(D.1) is exactly right.** Computing $\langle g_i^a(k) | H_g | 0 \rangle$ from (2.20) and (B.2) gives $g k^i p^a(-k) / (4\pi^{3/2} |k^+|^{\{3/2\}})$. ✓
- **(4.19) → (4.20) is right, and (4.20) → (5.2) is right:** matching (4.15)'s $(2\pi)^{\{3/2\}} \int d^3p / (2\pi)^3 A_2$ against (4.20) gives exactly $A_j^b(p) = -\sqrt{2} g p^j p^b(-p) / (\sqrt{(p^+)^2})$. ✓
- **(4.6) → (4.39) → (4.80) → (5.1) are mutually consistent.** $N^2 = 1 - \int A^\dagger A \Rightarrow N = 1 - \frac{1}{2} \int A^\dagger A$, and $\frac{1}{2} \int d^3p / (2\pi)^3 A^\dagger A = (g^2/8\pi^3) \log(v/\Lambda) \int d^2p p p / p^2$. ✓
- **(4.78) is right.** $\int_a^{\{1-a\}} d\xi [\xi(1-\xi) + (1-\xi)/\xi + \xi/(1-\xi)] = 1/6 - 2\ln a - 2 = -2\log(\Lambda/k^+) - 11/6$. ✓
- **(4.22) is right** — I initially thought the overall minus was spurious. It is not: matching (D.3)'s index assignment ($q \leftrightarrow l, p \leftrightarrow j$) onto (4.21)'s state $|g_j^b(k-p)g_k^c(p)\rangle$ flips the sign of all three tensor structures, which I checked term by term. $-igf^{\{abc\}}$ is correct, and (4.36)'s $+igf^{\{acb\}}$ is the same thing. ✓
- **(4.24) is right** (checked against D.4 including the $(2p^+ - k^+)$ factor). ✓
- **(4.53)–(4.54) are right:** the two orderings (4.50) and (4.52) combine with the decomposition (4.28), and the $(p^+k^2 - k^+p^2)$ denominators cancel exactly, leaving precisely (4.54). ✓
- **(6.9) is right:** $\int A^\dagger A p^2 / 2p^+ = \int g^2 p p / (p^+)^2$. ✓

§A. Definite errors

A1. Eq. (4.55) — missing factor i and missing $1/\sqrt{(k^+p^+)}$

This is the same calculation that produces (4.54), which is correct. Writing $X = \sqrt{(k^+)}/(\sqrt{(p^+)k^2})$, $Y = \sqrt{(p^+)}/(\sqrt{(k^+)p^2})$, the sum of (4.50) and (4.52) is

$$C \{ X p^b(-p) p^a(k) - Y p^a(k) p^b(-p) \}, \quad C = (g^2/4\pi^3) k^i p^j / (p^+k^2 - k^+p^2) \\ = C \{ \frac{1}{2}(X-Y) \{p, p\} + \frac{1}{2}(X+Y) [p^b(-p), p^a(k)] \}$$

- The $\{p, p\}$ piece gives $-(g^2/8\pi^3) k^i p^j / (\sqrt{(k^+p^+)k^2p^2}) \{p^a(k), p^b(-p)\} = \text{(4.54) exactly}$. ✓
- The $[p, p]$ piece gives

$$C \cdot \frac{1}{2}(X+Y) \cdot [p^b(-p), p^a(k)] \quad \text{with} \quad [p^b(-p), p^a(k)] = -i f^{\{abc\}} p^c(k-p) \\ \Rightarrow + i (g^2 f^{\{abc\}}/8\pi^3) \cdot k^i p^j (p^2k^+ + p^+k^2) / (\sqrt{(k^+p^+)} k^2 p^2 (p^2k^+ - p^+k^2)) p^c(k-p)$$

Eq. (4.55) as printed is

$$-(g^2 f^{\{abc\}}/8\pi^3) \int dp^+ \int d^2p |g_j^b(p)\rangle (k^i p^j / k^2 p^2) ((p^2k^+ + p^+k^2)/(p^2k^+ - p^+k^2)) p^c(k-p)$$

so it is missing (i) the factor i — which is mandatory, since (4.29) itself writes the commutator as $(i/2) f^{\{bad\}} p^d$; and (ii) the factor $1/\sqrt{(k^+p^+)}$ — note (4.54) has $\int dp^+ / \sqrt{(k^+p^+)}$ while (4.55) has bare $\int dp^+$, although both come from the same two terms.

A2. Eq. (5.4) — A1 propagates, producing a dimensionally inconsistent bracket

(5.4) contains

$$- (g^2/8\pi^3) (k^i p^j / k^2 p^2) [(1/\sqrt{(k^+p^+)}) \{p^a(k), p^b(-p)\} + f^{\{abc\}} p^c(k-p) (p^2k^+ + k^2p^+) / (p^2k^+ - k^2p^+)]$$

The two terms **inside the same bracket** differ by a factor $\sqrt{(k^+p^+)}$ and by a factor i . This is visible without any derivation: one term carries $1/\sqrt{(k^+p^+)}$, the other does not. The second term should read $+ i f^{abc} p^c (k-p) (p^2 k^+ + k^2 p^+) / (\sqrt{(k^+p^+)} (p^2 k^+ - k^2 p^+))$.

A3. Eq. (4.36) — the instantaneous contribution $|\Psi^3_{ggg}\rangle$ is missing

Eq. (4.35) states $|\Psi_{ggg}\rangle = |\Psi^{1a}_{gg[p,p]}\rangle + |\Psi^2_{ggg}\rangle + |\Psi^3_{ggg}\rangle$ — three pieces. Eq. (4.36) contains only two: the (4.31) term and the (4.22) term. The $|\Psi^3_{ggg}\rangle$ of (4.24) is absent.

This is not a harmless omission, because (5.3) **does include it** — its last term, $- i g^2 f^{abc} (p^+ - q^+) p^a (-p - q) \delta_{jk} / (2(2\pi)^3 \sqrt{(p^+q^+)} (p^+ + q^+)^2 (q^2/q^+ + p^2/p^+))$, is exactly (4.24) rewritten with $k \rightarrow p+q$ (I checked the $(2p^+ - k^+) \rightarrow (p^+ - q^+)$ substitution and the sign; they agree). So (4.36) is simply incomplete relative to what is actually used.

A4. Eq. (5.3) — stray momentum k in a function of (p, q)

In the third group of (5.3):

$$[2p^i - (2p^+/(p^++q^+)) (p^i + k^i)] \delta_{jk}$$

$B_2^{cb}_{kj}(p, q)$ depends only on p and q . Tracing back to (4.36) with $k \rightarrow p+q$ and $p \rightarrow q$, this term is $2p^i - (2p^+/(p^++q^+)) (p^i + q^i)$. $(p^i + k^i)$ **must be** $(p^i + q^i)$. Confirmed by the fact that the *other* two structures in the same bracket, and the trailing factor $g p^a (-p - q) (p^i + q^i) / (4\pi^{3/2} |p^+ + q^+|^{3/2})$, have already been correctly converted to p, q .

A5. Eq. (5.3) — momentum/index assignment is swapped relative to Eq. (4.1)

Eq. (4.1) defines $B_2^{cb}_{kj}(p, q)$ $a_j^{tb}(p)$ $a_k^{tc}(q)$: index pair (b, j) belongs to momentum p , and (c, k) to momentum q .

But every term of (5.3) uses the opposite assignment. The first two terms have q^j p^k $p^b(-q) p^c(-p)$ and p^k q^j $p^c(-p) p^b(-q)$ — index j on momentum q , index k on momentum p , colour b on q , colour c on p . The third group is the same (q^j sits with δ_{ik} , p^k with δ_{ij}). Checked against the source (4.38), where the assignment *is* correct (p^k with the gluon of momentum p and index k ; $(k-p)^j$ with the gluon $(k-p)$ of index j).

So (5.3) is really $B_2^{cb}_{kj}(q, p)$. Either swap $p \leftrightarrow q$ throughout (5.3), or write the arguments in the other order. (This is invisible after symmetrisation, but as written the equation does not match its own definition.)

A6. Eq. (4.86) — two colour/momentum errors, both fixed by comparison with (4.88)

(4.86) and (4.88) are the two time orderings of the same process (incoming $r(d, l)$, $s(e, m)$; outgoing $p(b, j)$, $q(c, k)$; s absorbed by the valence; $r \rightarrow p+q$).

- **Colour factor.** (4.86) has f^{dbe} . The splitting $r(d) \rightarrow p(b) + q(c)$ must give f^{dbc} , and the valence absorption supplies p^e . As printed, c is a free index absent from the colour tensor and e appears twice. (4.88) correctly has f^{dbc} . $f^{dbe} \rightarrow f^{dbc}$.
- **Argument of p .** (4.86) has $p^e(-s)$. The valence *absorbs* s , which by the paper's own convention (cf. (4.50), where absorbing k gives $p^a(k)$, and (4.44), where absorbing $k-p$ gives $p^c(k-p)$) is $p^e(s)$. (4.88) correctly has $p^e(s)$. $p^e(-s) \rightarrow p^e(s)$.

The rest of (4.86) — the s^m/s^2 from the $1/(s^2/2s^+)$ denominator, the $1/\sqrt{(p^+q^+r^+s^+)}$, and the three-term tensor structure — I checked and it is correct.

A7. Eq. (4.94) — g^4 should be g^2

$(\Psi^{ba}_{ji}(k, p))^3_{ggg}$ is built from **two** H_{ggg} insertions (4.93), i.e. $(ig)^2 = -g^2$. The printed prefactor is $-g^4 f^{ade} f^{bfh} / (128\pi^3)$. Compare (4.84), which is the same two- H_{ggg} topology and correctly carries $g^2 f^{d'ed} f^{d'cb} / (128\pi^3)$ — same $128\pi^3$, and g^2 . Also, this whole subsection is explicitly "at order g^2 ". $g^4 \rightarrow g^2$.

(The colour structure $f^{ade} f^{bfh}$ with no shared index *is* correct here — the diagram on p. 36 is two disconnected splittings, $k \rightarrow r, s$ and $p \rightarrow u, v$.)

A8. Eq. (6.4) — sign error; the printed bracket is anti-Hermitian and cannot cancel H_g

(6.4) prints

$$\int d^3p/(2\pi)^3 \left[-A_j^b(p) (p^2/2p^+) a_j^{\dagger b}(p) + A_j^{\dagger b}(p) (p^2/2p^+) a_j^b(p) \right]$$

Two independent arguments show the relative sign is wrong:

1. **Hermiticity.** $(A E a^\dagger)^\dagger = A^\dagger E a$, so the two terms are Hermitian conjugates of one another and must enter with the *same* sign. As printed the bracket is anti-Hermitian, while H_g — which it is supposed to cancel — is Hermitian. An anti-Hermitian operator cannot cancel a Hermitian one, so (6.5) cannot follow from (6.4) as written.
2. **Direct computation.** With $\Omega = 1 + g\Omega^{(1)}$, $\Omega^{(1)} = A a^\dagger - A^\dagger a$ (using (4.7)), and $\Omega^\dagger H_0 \Omega = H_0 + [H_0, \Omega^{(1)}]$:

$$[H_0, A a^\dagger - A^\dagger a] = +A (p^2/2p^+) a^\dagger + A^\dagger (p^2/2p^+) a$$

and with (5.2), $A_j^b(p) \cdot p^2/2p^+ = -g p^j p^b(-p)/(\sqrt{2} (p^+)^{3/2})$, which is *minus* the $a^\dagger$ coefficient of H_g in (2.20). So $H_g + [H_0, \Omega^{(1)}]|_A = 0$ **only** with $(+, +)$.

The first term should be $+ A_j^b(p) (p^2/2p^+) a_j^{\dagger b}(p)$.

For the record, the C -part of (6.4) *is* correct: the h.c. of $C_1(E_r+E_q-E_p)a^\dagger a^\dagger a$, written in the $a^\dagger a a$ basis, carries $(E_p+E_q-E_r) = -(E_r-E_q-E_p)$, which reproduces the printed $-C^\dagger(r^2/2r^+ - q^2/2q^+ - p^2/2p^+)$ exactly. Only the A terms are wrong.

A9. Eq. (2.23) — unbalanced brackets, duplicated prefactor, and a font break

As typeset, (2.23) opens $[$ after the first prefactor/measure, then in the middle repeats $+ (g^2/8) f^{abc} f^{ade} \int (dp^+/2\pi)(dk^+/2\pi)(dr^+/2\pi)dq^+ \int \dots 1/\sqrt{(p^+k^+r^+q^+)}$ and opens a **second** $[$, but only **one** $]$ ever closes. Two openings, one closing. Compare (2.24), which has the same two-group structure and *is* balanced ($[...] + \text{prefactor} \int [...]$). (2.23) is missing the closing bracket before the repeated prefactor.

In addition, the last operator product of (2.23), $a_i^{\dagger b}(k)a_j^c(p)a_i^{\dagger d}(r)a_j^{\dagger e}(q)$, is typeset in **upright roman** while every other operator in the equation is italic — a math-mode slip.

A10. Eqs. (4.97) and (4.99) — inconsistent integration measure

Both are three-gluon outgoing wave functions from three-gluon incoming states, and (4.100) adds them together. (4.97) integrates $\int d^3p d^3q d^3u$; (4.99) integrates $\int (d^3p/(2\pi)^3)(d^3q/(2\pi)^3)(d^3u/(2\pi)^3)$. Their sources (4.96) and (4.98) both use the bare d^3 measure. One of the two is wrong by $(2\pi)^9$.

A11. Duplicate bibliography entries

- **[22] and [34]** are the same paper: Altinoluk & Kovner, *Particle production at high energy and large transverse momentum: "The hybrid formalism" revisited*, PRD 83 (2011).
- **[47] and [50]** are the same paper: Munier, *Unitary perturbation theory on the light cone using adiabatic switching* ([50] additionally gives the eprint number 2510.05256).

§B. Internal inconsistencies to resolve

B1. Eq. (4.68) contradicts the paragraph immediately above it

The text on p. 29 states that "products mixing g and g^2 order components contribute only beyond the perturbative accuracy considered here and can therefore be neglected." It then singles out the overlap of $|(\Psi)_{gg}\rangle$ (order g , from H_{ggg}) with $|(\Psi)_{ggg}\rangle$ (order g^2) as "one potentially contributing term", and argues it vanishes "due to the symmetry properties". That overlap is $0(g^3)$ and is already excluded by the sentence above. Either drop (4.68) or explain why the earlier rule does not apply.

B2. Eq. (4.72)/(4.73) — one equation carrying two numbers, and broken cross-references

What is displayed is a single equation split across four lines, but it is given **two** numbers, (4.72) on the first line and (4.73) on the last. The following text then says "the contributions in the last line of Eq. (4.72)" and "The remaining contribution in Eq. (4.72)", but the last line is (4.73). Also, this display writes N_z in a calligraphic font where everywhere else the symbol is

blackboard-bold $\mathbb{N}_2$ (blackboard bold).

B3. B_3 is both matched and constrained, and the two are never reconciled

Eq. (4.10) determines $B_3^\dagger$ from the $O(g^2)$ unitarity condition, while (5.4) determines B_3 independently by matching to the LCWF. The text below (5.12) says only " B_1, D_3, E_3 are fixed by the constraint equation" — but B_3 appears in both places, so it is over-determined. That the two determinations agree is a genuine, nontrivial check of the whole construction. Either state that it has been performed, or say explicitly which route fixes B_3 and what (4.10) is then used for.

B4. Polarization index k collides with momentum k

(D.3) uses ℓ for the polarization index of the outgoing gluon q , giving $\delta_{j\ell}, \delta_{i\ell}, \delta_{ij}$ and q^ℓ, p^ℓ, k^ℓ — unambiguous. The main text renames it to k in (4.22), (4.36), (4.46), (4.48), (4.57) and (5.5), producing expressions such as

$$((k^+ + p^+) / q^+ \, q^\ell k - p^\ell k - k^\ell k) \, \delta_{ij} \quad [\text{Eq. (5.5)}]$$

where $k^\ell k$ means "component k of the momentum k ". This is not an error — I checked that the contractions are consistent — but it is genuinely ambiguous on first reading, and it is inconsistent with the appendix that the main text cites. Rename the polarization index to ℓ (or the momentum to something else) throughout.

B5. Index labels change between (4.23) and (4.24)

(4.23) writes the state $|g_i^b(k-p) \, g_j^c(p)\rangle$ and picks up δ_{ij} from (D.4); (4.24) writes the same state as $|g_j^b(k-p) \, g_k^c(p)\rangle$ with δ_{jk} . Harmless in isolation, but this is the kind of relabelling that propagates.

B6. Eq. (2.23) vs (2.24) — redundant and inconsistently labelled

(2.17) lists H_{gggg} and $H_{gggg-inst.}$ as separate terms, and (C.6) defines only their **sum**. (2.23) is labelled H_{gggg} alone, yet it contains momentum fractions such as $(p^+ + k^+) q^+ / ((r^+ + q^+) (k^+ - p^+))$ that can only come from the instantaneous $1/\partial^+$ structures — the genuine four-gluon contact vertex has no such factors. (2.24), labelled $H_{gggg} + H_{gggg-inst.}$, then covers overlapping ground. Given that (2.23) is also malformed (A9), my reading is that (2.23) is a leftover that should be deleted, with (2.24) retained.

B7. Non-normal-ordered operator products in (2.23)–(2.24)

Several products are written with annihilation operators to the left of creation operators, e.g. $a_i^b(k) \, a_i^{\dagger c}(p) \, a_j^{\dagger d}(r) \, a_j^{\dagger e}(q)$ in (2.23), and $a_i^b(p) \, a_i^{\dagger c}(k) \, a_j^{\dagger d}(r) \, a_j^{\dagger e}(q)$ in (2.24). Since $[a, a^\dagger] \neq 0$, these differ from their normal-ordered forms by one-body terms. Given that normal ordering is the organising principle of the entire paper, either normal-order these or state explicitly that the ordering as written is intended.

B8. Eq. (2.24), third term — squared longitudinal denominators

Terms 1 and 2 carry $1/((p^+ - k^+) (r^+ - q^+))$; term 3 carries $1/((p^+ + k^+)^2 (r^+ + q^+)^2)$. Each $1/\partial^+$ in (C.6) supplies one power, so the squares look anomalous relative to the neighbouring terms. Worth re-deriving.

§C. Conceptual points a referee will raise

C1. Eq. (6.7) → (6.8) is asserted, not shown — and it is the main result

The step from (6.6) to (6.7) drops H_{gggg} , $H_{gg-inst.}$ and $H_{gggg-inst.}$ with no intermediate algebra, and (6.8) asserts that the $C^\dagger C$ structure vanishes identically ("Evaluating the second term explicitly, one finds ... = 0") with no demonstration. This is the paper's central application. At minimum it needs an appendix: the $O(g^2)$ transformation involves $[H_0, \Omega^{(2)}]$, $[H_g + H_{ggg}, \Omega^{(1)}]$ and $\frac{1}{2} [[H_0, \Omega^{(1)}], \Omega^{(1)}]$, and a reader currently cannot verify that they have all been included, let alone that they cancel.

C2. The scaleless-integral argument on p. 30 needs justification in *this* scheme

$\langle (\Psi)_{gg} | (\Psi)_{gg} \rangle$ is set to zero because $\int d^2 p \sim / p^2$ is scaleless, citing dimensional regularisation [62]. But the calculation is not done in dimensional regularisation: the longitudinal directions carry explicit cutoffs Λ and ν , and the same transverse integral $\int d^2 p \, p^\alpha a(-p) p^\alpha a(p) / p^2$ is kept unintegrated elsewhere in the very same equation. Discarding an

overlapping UV/IR divergence by fiat, in a cutoff scheme, is a scheme choice that changes $\mathbb{N}$. The UV half of it would ordinarily be absorbed into coupling renormalisation, not dropped. State the scheme explicitly.

C3. E_1 and E_2 are pure products of C 's — so why are they in the generator G ?

(5.8) and (5.9) give E_1 and E_2 entirely as products $C \cdot C$ and $C \cdot C^\dagger$ over energy denominators, i.e. as disconnected/reducible structures. The text introducing (5.11) says G is introduced "to extract the connected contributions". But (5.12) then carries E_1 , E_2 , E_3 terms explicitly. If G is meant to hold only connected pieces, those products should cancel against the exponentiation of the C terms and should not appear in G . Either the "connected" language or the content of (5.12) needs adjusting.

C4. Λ - and v -dependence is raised and then dropped

§2.3 states correctly that "physical observables should ultimately be independent of the arbitrary separation scale v ". The coefficients themselves are manifestly cutoff-dependent — (5.4) carries $\theta(p^+ - k^+ - \Lambda)$ and $\theta(k^+ - \Lambda - p^+)$, and $\mathbb{N}$ in (4.80) is $\propto \log(v/\Lambda)$. Nothing in the paper shows or even sketches how the dependence cancels. This will be asked.

C5. (6.10): the surviving energy has no transverse propagator

$\int d^3p / (2\pi)^3 g^2 p^+ b(p) p^+ b(-p) / (p^+)^2$ — the $1/p^2$ of the Weizsäcker–Williams field has cancelled completely against the light-cone energy $p^2/2p^+$. The arithmetic is right (I checked (6.9)), but the $\int dp^+ / (p^+)^2$ is *power* divergent at the lower cutoff, not logarithmic. Since the text calls this "the coherent background field energy", one sentence on why a linear $1/\Lambda$ sensitivity is the expected answer would help the reader.

C6. LCPT citations are thin

"light-cone perturbation theory (LCPT) [46, 47]" cites Kovchegov–Levin's book and a 2025 Munier paper. The standard references (Lepage–Brodsky; Bjorken–Kogut–Soper; Brodsky–Pauli–Pinsky) are absent. Also, "As discussed in Appendix A of Ref. [61]" points at a *Physics Letters B* paper (Kovner–Lublinsky–Serino, PLB 792) — worth confirming that it has an Appendix A on adiabatic switching, since a Munier reference would be the natural home for that statement.

Finally, in the concluding paragraph, "...requires the explicit coefficients of the evolution operator derived here [63]" attaches [63] (Kovner–Lublinsky, *One gluon, two gluon*) to a clause about coefficients derived *in this paper*. The citation is misplaced or belongs to a different clause.

§D. Typography and prose

Hyphenated plurals — "two-gluons", "three-gluons" used as plural nouns, and "a single-gluon" used as a noun. These should be "two gluons", "three gluons", "a single gluon". Occurrences: p. 17 ("production of two-gluons"), p. 18 ("emission of two-gluons"), p. 25 ("splitting into two-gluons"), p. 26 ("splits into three-gluons"), p. 26 ("splits into two-gluons"), p. 27 ("splits into two-gluons"), p. 28 ("first splits into two-gluons"), p. 31 ("containing two-gluons"), p. 35 ("merge into a single-gluon"), p. 38 ("merge into a single-gluon").

Other prose:

Location	Printed	Should be
p. 38	"then an another incoming gluon splits"	"then another"
p. 30	"the contributions ... sums up to"	"sum up to"
p. 22	"we get identical expression modulo the integration limits"	"an identical expression"
p. 48 (App. B)	"the following three eigen states"	"eigenstates"
p. 47	"research assisstantship support"	"assistantship"
Fig. 1 caption	"the green region – $\Lambda < p^+ < v$ "	stray en-dash before Λ

Inconsistent hyphenation throughout (pick one and apply globally):

- "Born–Oppenheimer" (abstract, p. 2, p. 5) vs "Born-Oppenheimer" (contents, §2.3 heading, p. 5, p. 46)
- "non-commuting" (abstract) vs "non commuting" (p. 11, p. 11 again, p. 46)
- "off diagonal" (abstract, p. 10) vs "off-diagonal" elsewhere

- "next-to-leading" vs "next to leading" (p. 32, p. 38, p. 50)
- "multi particle" (p. 40) vs "multi-gluon" (p. 1, p. 2)
- "Weizsäcker-Williams" vs "Weizsäcker-Williams" (p. 45, p. 46)

Math typography:

- (5.3), term 2: $p^k (q)^j$ — the q is in plain italic and parenthesised, while every other transverse momentum in the paper is bold. Should be $p^k q^j$.
 - (2.23), last term: entire operator product in upright roman (see A9).
 - (2.23)/(2.24): $+ h.c.$, with a trailing comma inside the display, four times.
 - (5.4): $p^c(-p+k)$ on the first line vs $p^c(k-p)$ on the third — same object, two notations.
-

Priority order for fixing

1. **A8** (6.4 sign) — it breaks the paper's stated main result.
2. **A1 + A2** (4.55 and 5.4) — a coefficient of Ω is wrong by $i \cdot \sqrt{(k+p)}$.
3. **A3** (4.36 incomplete), **A4** (5.3 stray k), **A5** (5.3 argument order).
4. **A9 + B6** (2.23 malformed / redundant with 2.24).
5. **C1** (show the 6.6 $\rightarrow$ 6.8 algebra) — the largest referee-facing gap.
6. **A6, A7, A10, A11** — localized, quick.
7. **§B4** (index collision) and **§D** — presentation.